

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

MARK SCHEME

CLASS

INDEX NUMBER

CHEMISTRY

9729/01

Paper 1 Multiple Choice

XX XXX 2018

1 hour

Additional Materials: *Data Booklet*
Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your index number, name and class on all the work you hand in.
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

1	C	6	A	11	A	16	A	21	C	26	B
2	A	7	C	12	B	17	A	22	D	27	D
3	B	8	D	13	D	18	D	23	C	28	D
4	C	9	B	14	D	19	C	24	D	29	B
5	B	10	A	15	C	20	D	25	A	30	D

A: 7 B: 6 C:7 D: 10

This document consists of **X** printed pages.



Innova Junior College

[Turn over

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 In an experiment, 0.002 mol acidified iron (II) sulfate is quantitatively oxidised by 0.001 mol of an oxidising agent, ZO_3^- . Given that Fe^{2+} is oxidised to Fe^{3+} , what is the final oxidation state of **Z**?

A +1 **B** +2 **C** +3 **D** +4

Answer: C

Amount of electrons released by Fe^{2+} = 0.002 mol

= amount of electrons accepted by YO_3^-

Amount of YO_3^- ; amount of e = 1 : 2

Oxidation number of Y in YO_3^- = +5

New oxidation number of Y = +5 – 2 = +3

- 2 Gaseous particle **X** has a proton number n and a charge of +1.

Gaseous particle **Y** has a proton number of $(n + 1)$ and is isoelectronic with **X**.

Which of the following statements correctly describe **X** and **Y**?

- 1** **Y** has a smaller radius than **X**.
2 **Y** requires less energy than **X** when a further electron is removed from each particle.
3 **Y** releases less energy than **X** when an electron is added to each particle.

- A** 1 only
B 2 only
C 1 and 2 only
D 2 and 3 only

Answer: A

particle	proton number	electron number	charge
X	n	$n-1$	+1
Y	$n+1$	$n-1$	+2

Since **Y** has higher nuclear charge than **X**, and both **X** and **Y** have same shielding effect (same electron number), **Y** has higher effective nuclear charge and valence electrons are more strongly attracted to the nucleus.

Thus, **Y** has smaller radius than **X**. (Option 1 is correct)

Y requires more energy than **X** when a further electron is removed from each particle. (Option 2 is wrong)

Y releases more energy than **X** when an electron is added to each particle. (Option 3 is wrong)

- 3 What is the electronic configuration of an element with a **second** ionisation energy higher than that of each of the elements either side of it in the Periodic Table?

- A $1s^2 2s^2 2p^6 3s^2$
B $1s^2 2s^2 2p^6 3s^2 3p^1$
 C $1s^2 2s^2 2p^6 3s^2 3p^2$
 D $1s^2 2s^2 2p^6 3s^2 3p^3$

Answer: B

The electronic configuration for second ionisation energy of the above species:

- A: $1s^2 2s^2 2p^6 3s^1$
 B: $1s^2 2s^2 2p^6 3s^2$
 C: $1s^2 2s^2 2p^6 3s^2 3p^1$
 D: $1s^2 2s^2 2p^6 3s^2 3p^2$

As there is a decrease in ionisation energy between the species between ns^2 and $ns^2 np^1$, species that have electronic configuration of ns^2 will have second ionisation energy higher than that of each of the elements either side of it (ns^1 and $ns^2 np^1$) in the Periodic Table.

- 4 The table shows the charge and radius of each of six ions.

ion	J^+	L^+	M^{2+}	X^-	Y^-	Z^{2-}
radius / nm	0.14	0.18	0.15	0.14	0.18	0.15

The ionic solids JX, LY and MZ are of the same lattice type. What is the correct order of their lattice energies, placing the most exothermic first?

- A JX, MZ, LY
 B LY, MZ, JX
C MZ, JX, LY
 D MZ, LY, JX

Answer: C

Since $\left| \text{Lattice energy, } \Delta H_{\text{latt}}^\theta \right| \propto \left| \frac{q_+ q_-}{r_+ + r_-} \right|$, compound MZ has the most exothermic lattice

energy as M^{2+} and Z^{2-} has higher charge than that of J^+ , L^+ , X^- and Y^- . Compound LY has the least exothermic lattice energy as L^+ and Y^- has larger ionic radius than that of J^+ and Y^- .

- 5 In certain microwave ovens, the wave energy produced is absorbed by certain polar molecules.

Which of the following would absorb microwave energy?

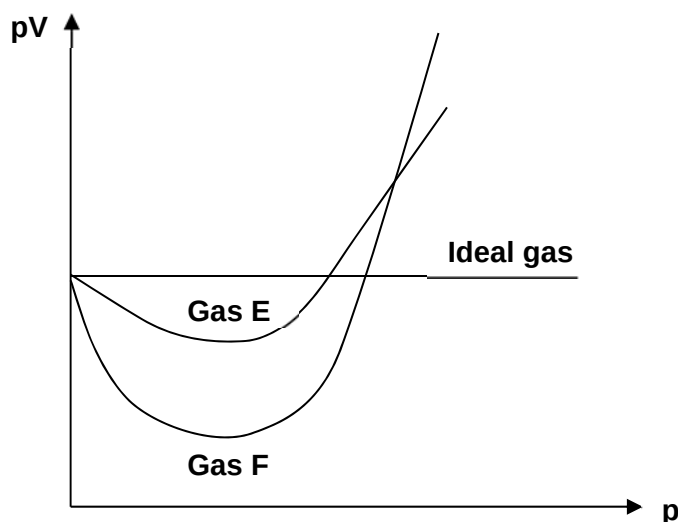
- 1 $\text{CH}_3\text{CH}_2\text{OH}$
 - 2 AlCl_3
 - 3 CO_2
 - 4 CH_3F
- A 1 and 2 only
- B** 1 and 4 only
- C 2 and 3 only
- D 3 and 4 only

Answer: B

Option 2 is non polar molecule. AlCl_3 is trigonal planar in shape and there is no net dipole moment in the molecule as all the dipole moments cancel out each other.

Option 3 is non polar molecule. CO_2 is linear in shape and there is no net dipole moment in the molecule as all the dipole moments cancel out each other.

Option 1 and 4 are polar as there is net dipole moment in the molecule.



Which one of the following statements is true for the observation?

- A** Gas E could be He while gas F could be CO₂.
- B** Gas E could be NH₃ while gas F could be H₂.
- C** Gas E could form intermolecular hydrogen bonds while gas F could form intermolecular permanent dipole-permanent dipole.
- D** Gas E could form stronger intermolecular instantaneous dipole-induced dipole than gas F.

Answer: A

$$pV = nRT$$

$$1.5 \times 10^{25} \times V = 1 \times 8.31 \times (273+27)$$

$$V = 0.0164 \text{ m}^3$$

$$V = 16.4 \text{ dm}^3$$

From the above calculation, it can be deduced that gas E behave more ideally than gas F.

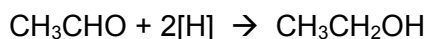
Option A is correct. He behaves more like an ideal gas as it has smaller electron cloud size and weaker i.d.-i.d. than that of CO₂. Intermolecular forces of attraction between He is negligible.

Option B is wrong. NH₃ can form hydrogen bonds between molecules and there is significant intermolecular forces of attraction. NH₃ deviates more from ideal gas.

Option C is wrong. Gas E should form weaker i.d.-i.d. between molecules than that of F as gas E behave more ideally than gas F.

Option D is wrong. Gas E should form weaker i.d.-i.d. than gas F as gas E behave more ideally than gas F.

- 7 The enzyme alcohol dehydrogenase catalyses an important step in the production of ethanol by fermentation.



The graph shows how the rate of this enzyme-catalysed reaction varies with the concentration of ethanal.

[ethanal]

Which statement best explains the reason for the flattening off of the curve?

- A All the ethanal has been used up and the reaction has finished.
- B As the ethanol product builds up the reaction slows down.
- C** At high ethanal concentrations all the active sites in the enzyme molecules are occupied by ethanal molecules.
- D At high ethanal concentrations the ethanal inhibits the action of enzyme.

Answer: C

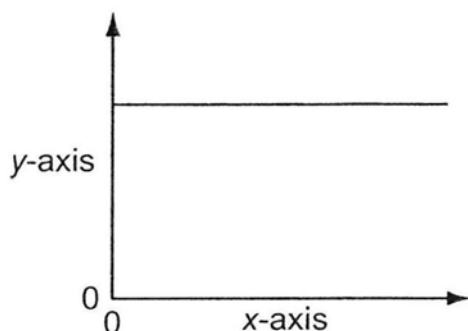
Option A is wrong as there is an increase in the concentration of ethanal. The ethanal has not been used up and the reaction has not finished.

Option B is wrong as the concentration of the ethanol product builds up the reaction will not slow down the rate of reaction.

Option D is wrong as ethanal is not an inhibitor but substrate (reactant) of this reaction.

- 8 The kinetics of the zero order reaction $P \rightarrow Q$ were investigated under different conditions. The table shows pairs of quantities that were plotted as graphs.

Which pairs gave the following graph?



	y-axis	x-axis
1	rate	time
2	rate constant	time
3	rate constant	temperature
4	rate	concentration

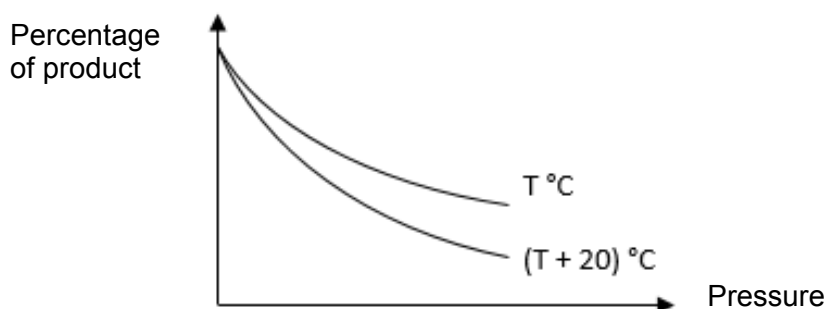
- A 1 and 4 only
 B 2 and 3 only
 C 1, 2 and 3 only
D 1, 2 and 4 only

Answer: D

For a zero order reaction, rate of the reaction is independent of time and concentration.

Also, rate constant is dependent on temperature and activation energy (from Arrhenius equation.)

- 9 The graph below shows how the percentage of product present at equilibrium varies with temperature and pressure for a reaction.



Which reaction could the graph represent?

- A $4\text{Fe (s)} + 3\text{O}_2\text{ (g)} \rightleftharpoons 2\text{Fe}_2\text{O}_3\text{ (s)}$ $\Delta H = -1644 \text{ kJ mol}^{-1}$
- B** $2\text{C (s)} + \text{O}_2\text{ (g)} \rightleftharpoons 2\text{CO (g)}$ $\Delta H = -222 \text{ kJ mol}^{-1}$
- C $\text{N}_2\text{O}_4\text{ (g)} \rightleftharpoons 2\text{NO}_2\text{ (g)}$ $\Delta H = +57.2 \text{ kJ mol}^{-1}$
- D $\text{CO (g)} + \text{Cl}_2\text{ (g)} \rightleftharpoons \text{COCl}_2\text{ (s)}$ $\Delta H = +86 \text{ kJ mol}^{-1}$

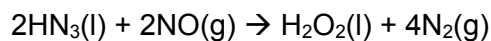
Answer: B

From the graph, when the pressure increases, the proportion of product molecules decrease. This indicates that when pressure increases, the equilibrium position shifts to left (which have fewer gaseous molecules to decrease the pressure). Option A and option D is wrong.

From the graph, when the temperature increases, the proportion of product molecules decrease too. This indicates that when temperature increases, the equilibrium position shifts to left (which is endothermic to reduce the temperature. Since option B is exothermic in the forward reaction, it would be endothermic in the backward reaction.

Only Option B shows the correct trend when temperature and pressure increase.

- 10 The standard enthalpy change of the following reaction is $-896.4 \text{ kJ mol}^{-1}$.



	NO(g)	H ₂ O ₂ (l)
$\Delta H_f^\theta / \text{kJ mol}^{-1}$	+ 90.3	- 187.8

Using the standard enthalpy change of formation in the above table, what is the standard enthalpy change of formation, in kJ mol^{-1} , of $\text{HN}_3(\text{l})$?

- A** +264
B +528
C +618
D +632

Answer: A

$$\text{Using } \Delta H_r^\theta = \sum n \Delta H_f^\theta (\text{products}) - \sum m \Delta H_f^\theta (\text{reactants})$$

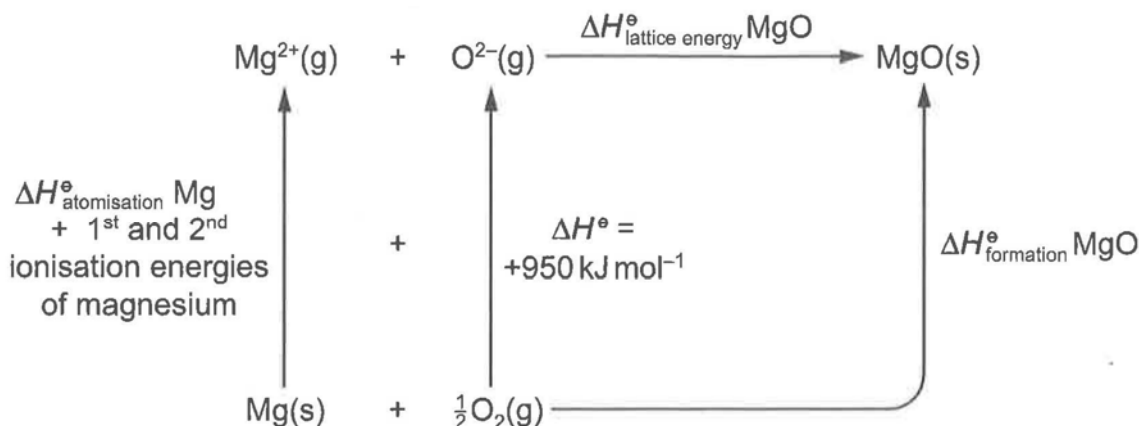
$$\Delta H_r^\theta = \Delta H_f (\text{H}_2\text{O}_2) - [2 \Delta H_f (\text{HN}_3) + 2 \Delta H_f (\text{NO})]$$

$$-896.4 = -187.8 - (2 \Delta H_f (\text{HN}_3) + 2 \times 90.3)$$

$$\Delta H_f (\text{HN}_3) = +264 \text{ kJ mol}^{-1}$$

- 11 A student calculated the lattice energy for magnesium oxide using the cycle shown.

The enthalpy change of atomisation of magnesium, $\Delta H_{\text{atomisation}}^{\theta} \text{Mg}$, is the energy needed when 1 mol of gaseous magnesium atoms is formed from 1 mol of solid magnesium under standard conditions.



However, the value calculated by the student for the lattice energy was **less** exothermic than the correct value.

Which errors could have been made in the calculation?

- 1 omitting the enthalpy change of atomisation of magnesium
- 2 omitting the first ionisation energy of magnesium
- 3 using the standard enthalpy change of combustion of magnesium rather than the standard enthalpy change of formation of magnesium oxide

- A** 1 and 2 only
B 1 and 3 only
C 2 and 3 only
D 1, 2 and 3

Ans: A

By Hess's law,

$$\text{Lattice energy} = - [\Delta H_{\text{atomisation}}^{\theta} \text{Mg}] - [1^{\text{st}} \text{ and } 2^{\text{nd}} \text{ I.E. of Mg}] - 950 + \Delta H_{\text{formation}}^{\theta} \text{MgO}$$

1 is correct as the lattice energy will be less exothermic if the enthalpy change of atomisation of Mg is omitted. This is because, the enthalpy change of atomisation of Mg is a positive term.

2 is correct as the lattice energy will be less exothermic if the first ionisation energy of Mg is omitted. This because, the first ionisation energy of Mg is a positive term

3 is incorrect as the replacing the standard enthalpy change of formation of MgO with the standard enthalpy change of combustion of Mg will have no effect on the value of the lattice energy. This is because both values are the same.

12 Which changes are accompanied by an increase in entropy of the system?

- 1 freezing of water
- 2 sublimation of iodine
- 3 vaporisation of ethanol
- 4 conversion of $O_2(g)$ to $O_3(g)$

- A 1 and 2 only
B 2 and 3 only
 C 3 and 4 only
 D 1 and 4 only

Answer: B

For option 2 and 3, there is an increase in the number of gaseous molecules (from solid to gas and from liquid to gas), there is an increase in the disorderness of the system, entropy increases.

For option 1, there is a decrease in the disorderness of the system (from liquid to solid) and similarly for that of option 4. $3O_2(g) \rightarrow 2O_3(g)$

13 The value of the ionic product of water, K_w , varies with temperature.

Temperature / °C	$K_w / \text{mol}^2 \text{dm}^{-6}$
25	1.0×10^{-14}
62	1.0×10^{-13}

What can be deduced from this information?

- A The ionic dissociation of water is an exothermic process.
 B The association of water molecules by hydrogen bonding increases as temperature increases.
 C The pH of pure water increases with temperature.
D At 62 °C, water with a pH of 6.5 is neutral.

Answer: D

Option A is wrong as ionic dissociation of water involves bond breaking of water molecules, this is an endothermic reaction.

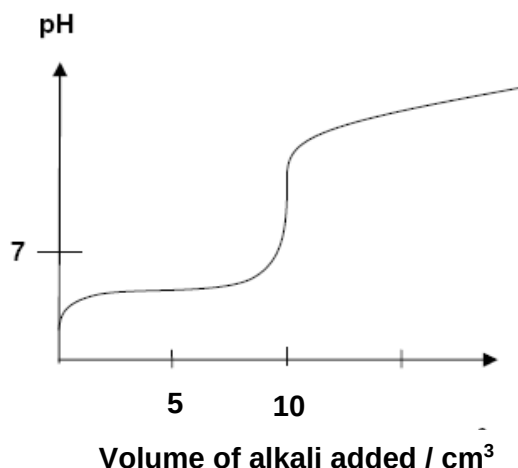
Option B is wrong. As temperature increases, more hydrogen bonds between molecules are broken.

Option C is wrong. The pH of pure water decreases with temperature. (pH = 7 at 25 °C and pH = 6.5 at 62 °C)

Option D is correct as water is neutral at all temperature as $[H^+] = [OH^-]$.

- 14 The following graph shows the changes in pH of 20.0 cm³ of 1.0 mol dm⁻³ acid solution when excess 1.0 mol dm⁻³ alkali solution is added gradually.

Which one of the following pairs of solution with a suitable indicator could have resulted in the graph below?



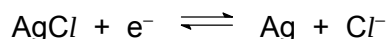
	alkali	acid	indicator
A	NaOH	H ₂ SO ₄	Methyl orange
B	NH ₃	HNO ₃	Phenolphthalein
C	KOH	CH ₃ CO ₂ H	Methyl orange
D	Ba(OH) ₂	CH ₃ CH ₂ CO ₂ H	Phenolphthalein

Answer: D

Since the volume of alkali needed for neutralisation is half of that of acid, this indicates that there should be 2 OH⁻ per base molecules. Since this is a weak acid-strong base titration, pH at equivalence point should be greater than 7 as a basic salt is formed. Phenolphthalein is a suitable indicator as its working range is greater than 7.

- 15 Use of the Data Booklet is relevant to this question.

The AgCl / Ag reference electrode is based on the following reaction:



The reduction potential of the Fe²⁺ / Fe half-cell was found to be -0.64 V when measured using the AgCl / Ag reference electrode.

What is the reduction potential of the AgCl / Ag electrode with respect to the standard hydrogen electrode?

- A -1.10 V B -0.20 V C +0.20 V D +1.10 V

Answer: C

The reduction potential of the Fe²⁺ / Fe half-cell was found to be -0.44 V when measured using the standard hydrogen electrode.

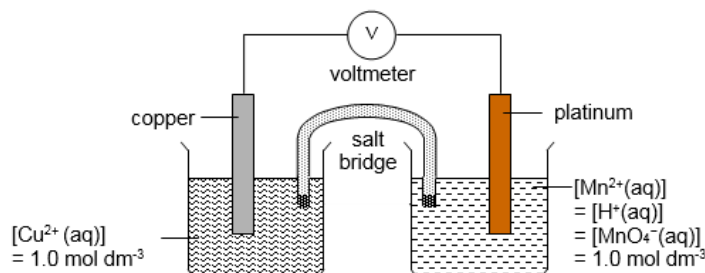
$$E_{\text{cell}}^{\circ} = E_{\text{red}}^{\circ} - E_{\text{ox}}^{\circ}$$

$$-0.44 \text{ V} = E_{\text{red}}^{\circ} - 0.64$$

$$E_{\text{red}}^{\circ} = +0.20 \text{ V}$$

16 Use of the Data Booklet is relevant to this question.

A cell is set up by connecting a Cu^{2+}/Cu half-cell and an acidified $\text{MnO}_4^-/\text{Mn}^{2+}$ half-cell.



Which change in the set up will cause the corresponding effect on the e.m.f (potential difference) of the cell?

	change	effect on e.m.f of cell
A	addition of excess NH_3 (aq) into oxidation half-cell	increases
B	addition of concentrated H_2SO_4 into reduction half-cell	decreases
C	increase the size of the copper electrode	increases
D	replace copper with an alloy of copper and zinc	remains the same

Answer: A



$$E^\circ_{\text{cell}} = E^\circ_{\text{red}} - E^\circ_{\text{ox}} = +1.52 - (+0.34) = +1.18 \text{ V}$$

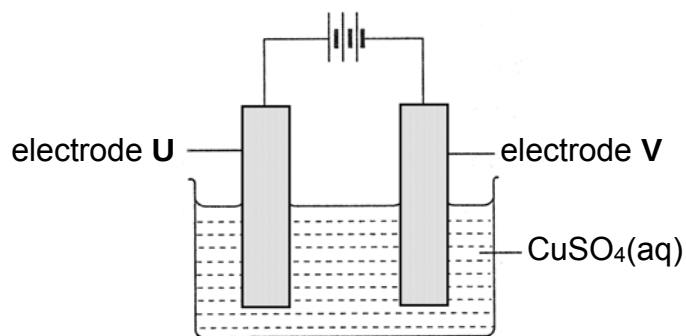
Option A is correct: when excess NH_3 (aq) into oxidation half-cell, $[\text{Cu}^{2+}]$ decreases as Cu^{2+} reacts with NH_3 to form $[\text{Cu}(\text{NH}_3)_4]^{2+}$. Since $[\text{Cu}^{2+}]$ decreases, equilibrium position shifts to the left, E°_{ox} is less positive, E°_{cell} becomes more positive.

Option B is wrong. When concentrated H_2SO_4 is added into reduction half-cell, $[\text{H}^+]$ increases. Since $[\text{H}^+]$ increases, equilibrium position shifts to the right, E°_{red} is more positive, E°_{cell} becomes more positive.

Option C is wrong. Increase the size of the copper electrode will not have effect on e.m.f. of the cell.

Option D is wrong. Replace copper with an alloy of copper and zinc will result in more positive E°_{cell} . $(+1.52 - (-0.76)) = +2.28 \text{ V}$

- 17 In an experiment, a cell was set up to obtain pure copper from a copper-silver alloy as shown below. Electrode **V** is pure copper and electrode **U** is the copper-silver alloy.



When a current of 40.0 A flows through the electrolyte for 26.8 minutes, the mass of the anode changes by 26.47 g.

What is the percentage of silver by mass in the electrode **U**?

- A** 20%
- B** 40%
- C** 60%
- D** 80%

Answer: A

$$I \times t = n_e \times F$$

$$40 \times 26.8 \times 60 = n_e \times 96500$$

$$n_e = 0.6665 \text{ mol}$$

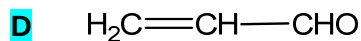
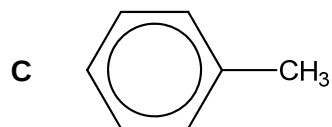
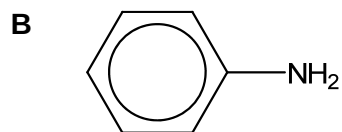
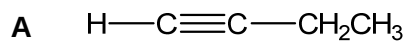
Since $\text{Cu} : e = 1 : 2$,

$$\text{Amount of Cu formed} = \frac{1}{2} \times 0.6665 = 0.3332 \text{ mol}$$

$$\text{Mass of Cu formed} = 0.3332 \times 63.5 = 21.16 \text{ g}$$

$$\% \text{ silver by mass} = (26.47 - 21.16) / 26.47 \times 100\% = 20\%$$

18 Which of the following is a planar molecule?



Answer: D

Option A is wrong. There are 2 sp^3 carbon atoms (tetrahedral around the C) in the molecule.

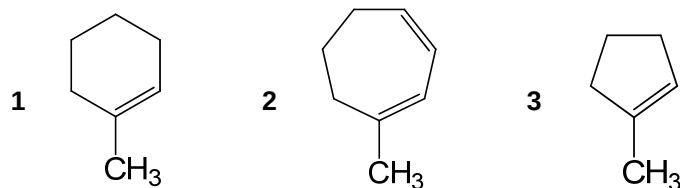
Option B is wrong. The molecule is trigonal pyramidal around the N atom.

Option C is wrong. There are 1 sp^3 carbon atom (tetrahedral around the C) in the molecule.

Option D is correct. All the carbons are sp^2 carbon in the molecule.

- 19 A hydrocarbon, on heating with an excess of hot concentrated acidic KMnO_4 produces $\text{HO}_2\text{CCH}_2\text{CH}_2\text{CH}_2\text{COCH}_3$.

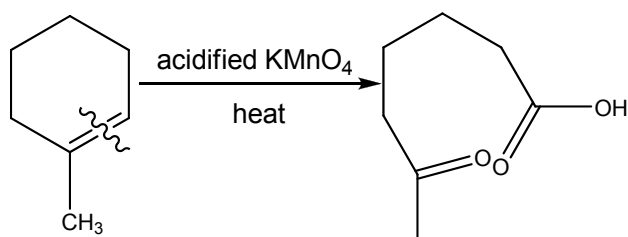
What could the hydrocarbon be?



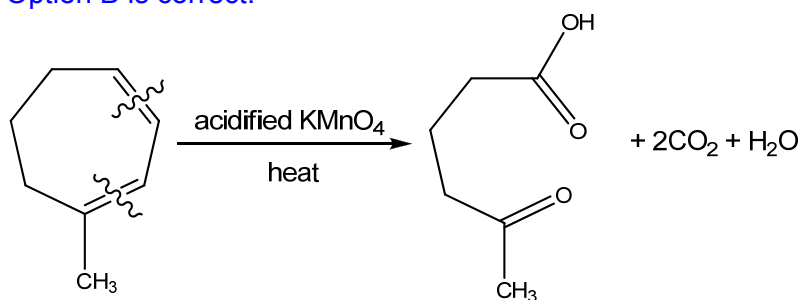
- A 1 and 2 only
 B 1 and 3 only
C 2 and 3 only
 D 1, 2 and 3

Answer: C

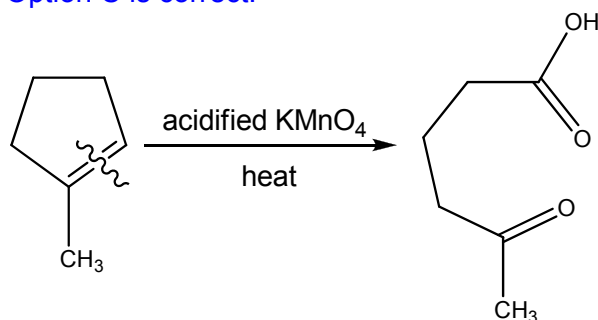
Option A is wrong.



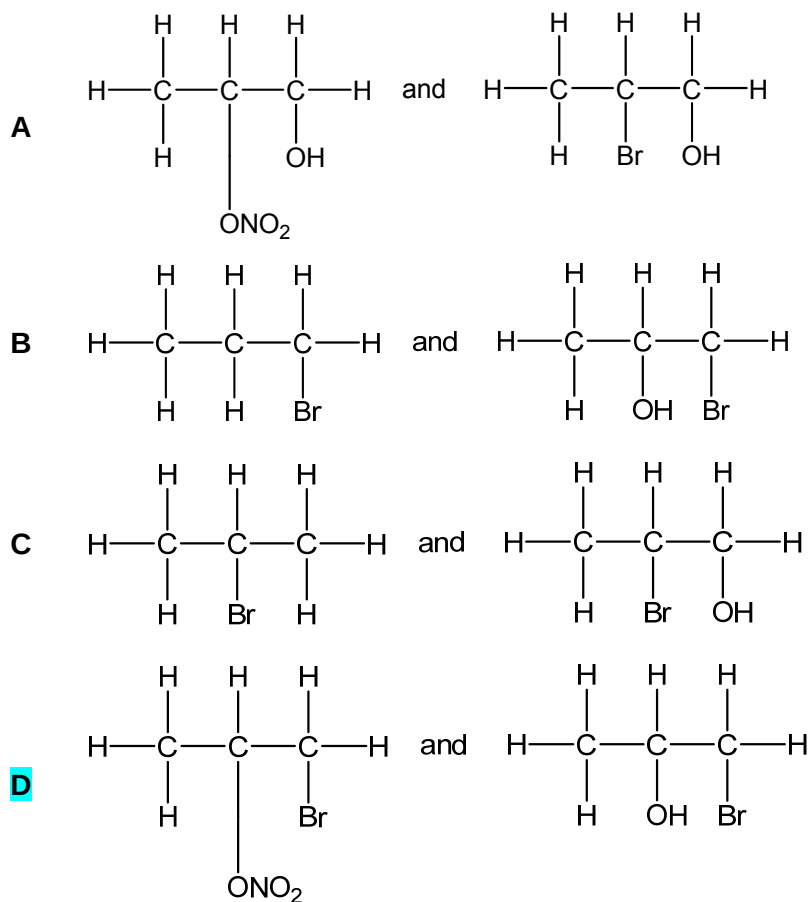
Option B is correct.



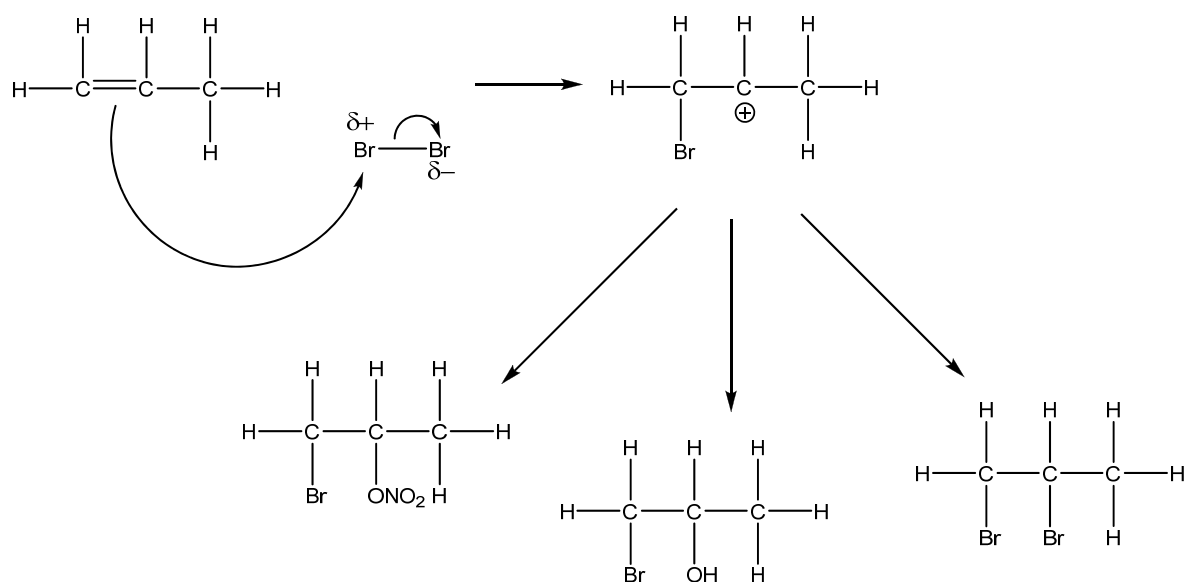
Option C is correct.



- 20 When propene reacts with Br_2 in the presence of excess aqueous KNO_3 , what are the two major products formed?



Answer: D



- 21 The molecule of benzene, C_6H_6 is a regular hexagon in which the π electrons are described as delocalised.

Which of the following statements are true for benzene?

- 1 It prevents benzene from undergoing addition reactions.
 - 2 All C–C bonds lengths in benzene are intermediate between C–C bond in an alkane and C=C bond in an alkene.
 - 3 The enthalpy change of hydrogenation of benzene is less exothermic than that predicted for cyclohexatriene.
 - 4 The presence of π electrons enables benzene to be a good electrical conductor.
- A 1 and 2 only
B 2 and 3 only
C 1, 2 and 3 only
D 1, 2 and 4 only

Answer: C

Option 1 is correct. Benzene undergoes substitution reaction to remain its aromaticity.

Option 2 is correct. As the π electrons are delocalised in the benzene ring, all the C-C bonds have partial double bond character.

Option 3 is correct. As the π electrons are delocalised in the benzene ring, benzene is more stable and it occupies lower energy level as it is more stable due to its aromaticity. As a result, less energy is released from hydrogenation of benzene.

Option 4 is wrong. Although the π electrons are delocalised in the benzene ring, there is no free mobile electrons and ions in liquid benzene, thus benzene is not a good conductor of electricity.

- 22 Some chlorobutanes were separately treated with hot ethanolic sodium hydroxide. Two of these gave the same hydrocarbon, C_4H_6 .

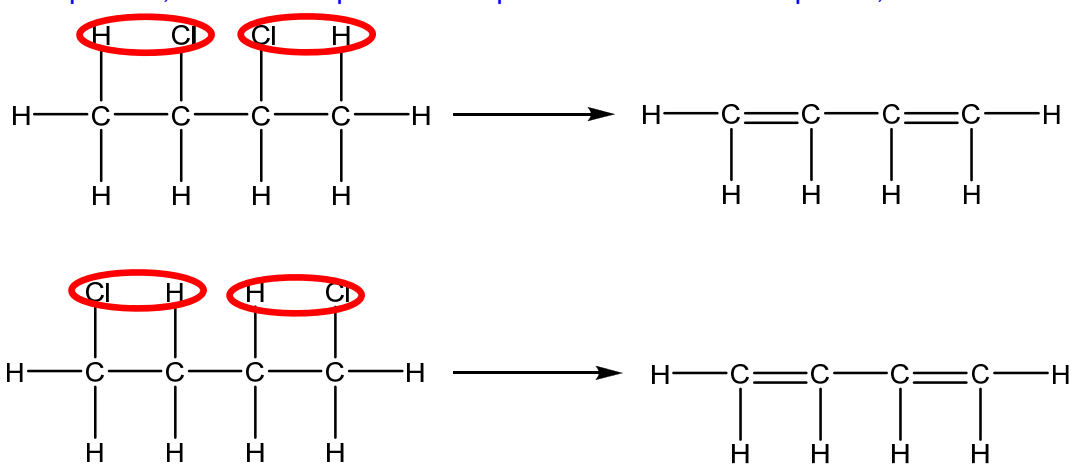
From which pair of chlorobutanes was this hydrocarbon obtained?

- A $CH_3CH_2CH_2CH_2Cl$ and $CH_3CH_2CH_2CHCl_2$
 B $CH_3CH_2CH_2CH_2Cl$ and $ClCH_2CH_2CH_2CH_2Cl$
 C $CH_3CH_2CCl_2CH_3$ and $ClCH_2CH_2CH_2CHCl_2$
D $CH_3CHClCHClCH_3$ and $ClCH_2CH_2CH_2CH_2Cl$

Answer: D

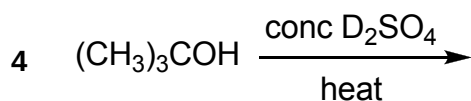
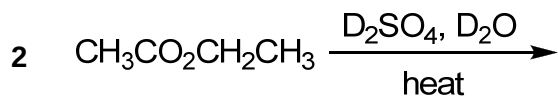
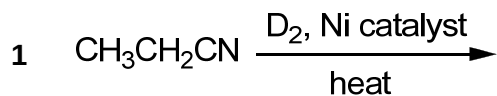
For option A, B and C, there is at least one molecule in the option contain 1 or 3 Cl atoms which after elimination can only form 1 or 3 C=C bond.

For option D, the two compounds can produce the same compound, as shown below:



23 Deuterium, D, is the ${}^2_1\text{H}$ isotope of hydrogen.

Which of the following reactions yield a carbon compound containing deuterium?



A 1 and 2 only

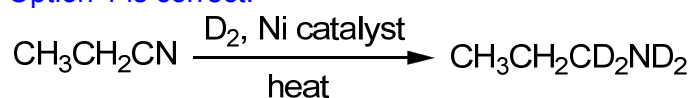
B 1 and 4 only

C 1, 2 and 3 only

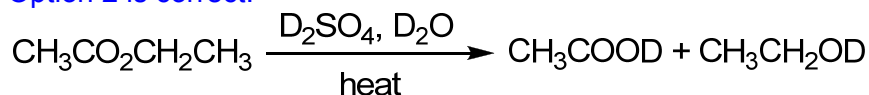
D 2, 3 and 4 only

Answer: C

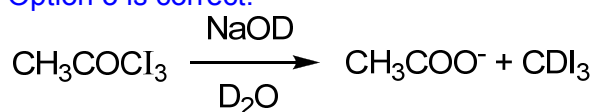
Option 1 is correct.



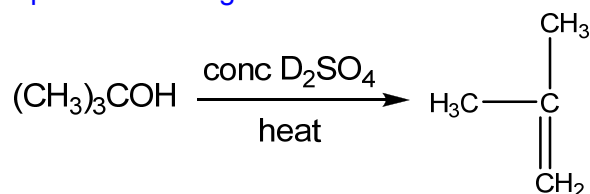
Option 2 is correct.



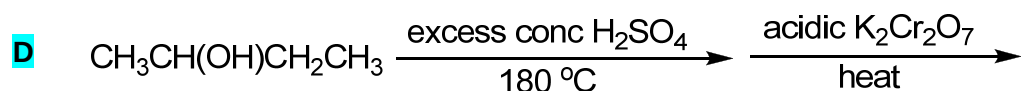
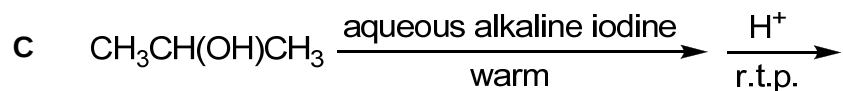
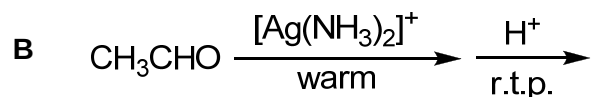
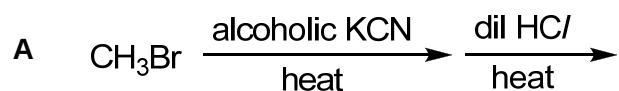
Option 3 is correct.



Option 4 is wrong.

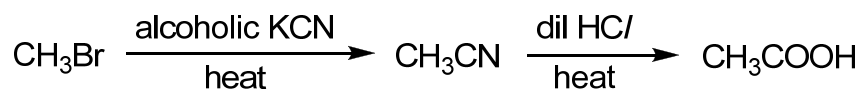


24 Which of the following synthetic routes **does not** produce ethanoic acid?

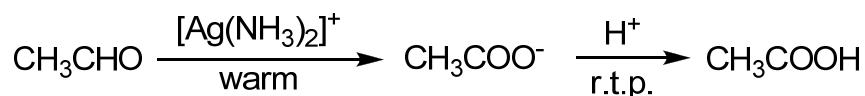


Answer: D

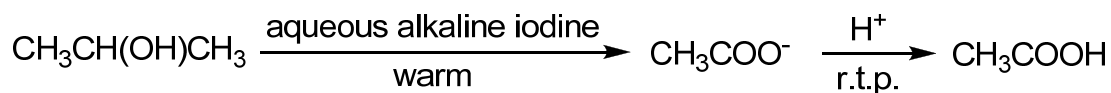
Option A is correct.



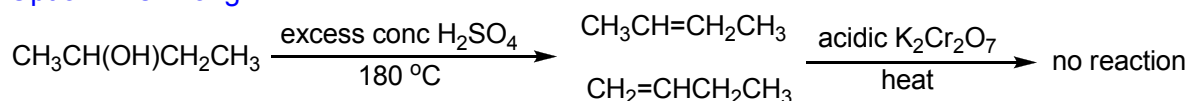
Option B is correct.



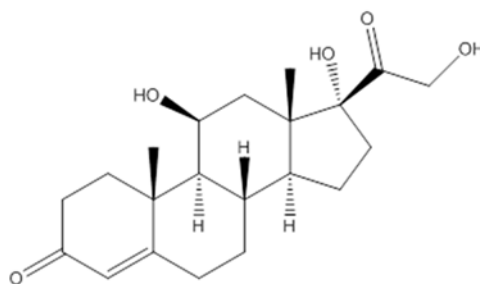
Option C is correct.



Option D is wrong.



- 25 Hydrocortisone is a steroid hormone produced by the adrenal gland and is released in response to stress. It is commonly used as an active ingredient in anti-inflammatory creams.



Hydrocortisone

Which of the following statements about hydrocortisone is true?

- A** When treated with NaBH_4 in the presence of methanol, it forms a compound containing 5 hydroxy groups.
- B** When warmed with aqueous alkaline iodine, a yellow precipitate is observed.
- C** When treated with cold dilute KMnO_4 , it forms a compound containing 2 hydroxy groups.
- D** When treated with an excess of hot concentrated acidified KMnO_4 , it forms a compound containing 7 chiral centres.

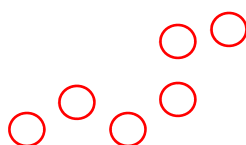
Answer: A

Option A is correct. The two ketone is reduced to secondary alcohol. Together with the other three alcohol in the molecule, the molecule has 5 hydroxy group.

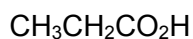
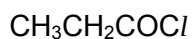
Option B is wrong. There is no $\text{H} / \text{R} - \text{C}(=\text{O}) - \text{CH}_3$ or $\begin{array}{c} \text{H} \\ | \\ \text{R} / \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_3 \end{array}$ in the molecule.

Option C is wrong. When treated with cold dilute KMnO_4 , the alkene in the molecule will be converted to diol. Together with the other three alcohol in the molecule, the molecule has 5 hydroxy group.

Option D is wrong. When treated with an excess of hot concentrated acidified KMnO_4 , it forms a compound containing 6 chiral centres.



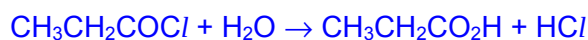
- 26 Equal amounts of compounds **P**, **Q**, **R** and **S** are separately shaken with 100 cm³ of water. The pH of each resultant solution is then measured.

**P****Q****R****S**

Which of the following shows the correct order of increasing order of acidity of the solutions formed?

A P, R, S, Q**B** P, S, R, Q**C** Q, P, S, R**D** Q, S, R, P

Answer: B

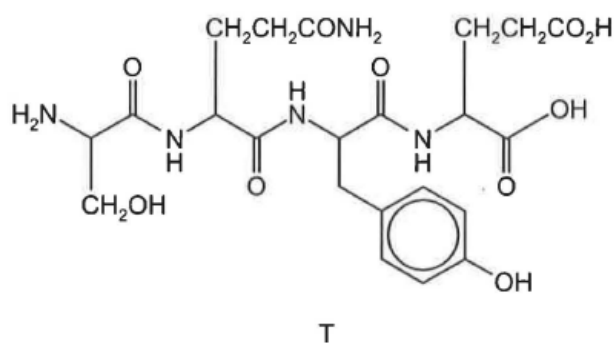


Solution is most acidic.

Cl group is more electronegative than Br and the negative charge of $\text{ClCH}_2\text{CH}_2\text{CO}_2^-$ is dispersed to a greater extent than $\text{BrCH}_2\text{CH}_2\text{CO}_2^-$. Thus, $\text{ClCH}_2\text{CH}_2\text{CO}_2^-$ is more stable than $\text{BrCH}_2\text{CH}_2\text{CO}_2^-$. $\text{ClCH}_2\text{CH}_2\text{CO}_2\text{H}$ is more acidic than $\text{BrCH}_2\text{CH}_2\text{CO}_2\text{H}$.

$\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$ is the least acidic among the four compounds.

27 The diagram shows the structure of the tetrapeptide **T**.



When 0.1 mole of **T** is heated under reflux with NaOH(aq) until no further reaction occurs, how many moles of NaOH will react?

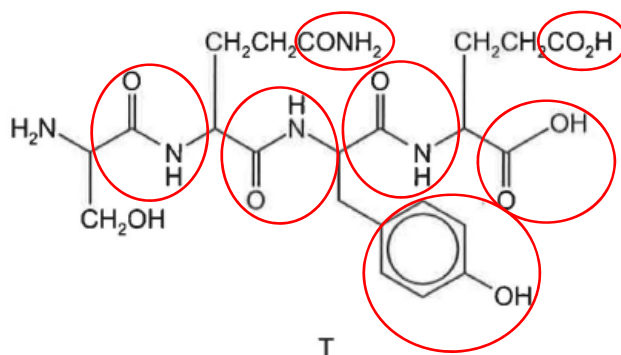
A 0.4

B 0.5

C 0.6

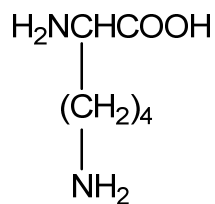
D 0.7

Answer: D

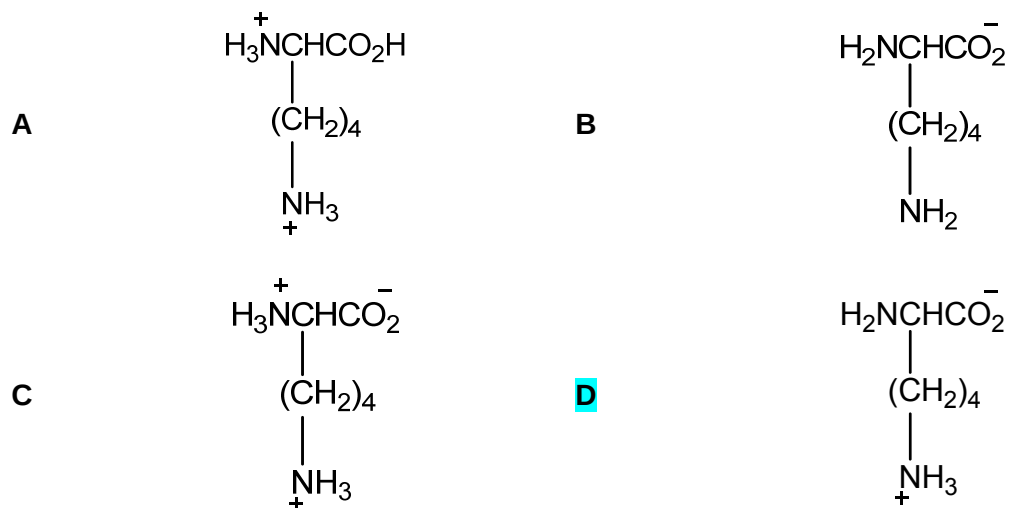


The circled functional group can react with NaOH(aq), heat with reflux.

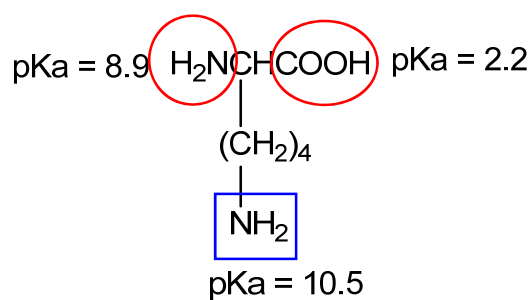
28 Lysine is an α -amino acid.



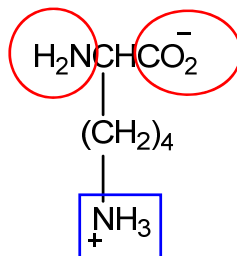
Which structure is predominant when lysine is in an aqueous solution of pH 9.5, given that lysine has three pK_a values of 2.2, 8.9 and 10.5?



Answer: D



When $\text{pH} > \text{pK}_a$, the functional group circled will be deprotonated.
 When $\text{pH} < \text{pK}_a$, the functional group squared will be protonated.



29 Element **J** is in Period 3 of the Periodic Table. The following three statements are the properties of element **J** or its compounds.

- Adding NaOH(aq) to the solution resulting from the reaction of a chloride of **J** with water produces a white precipitate which is soluble in an excess of NaOH(aq).
- Element **J** is a solid at room temperature.
- The oxide of element **J** is soluble in hydrochloric acid.

What is the identity of element **J**?

A magnesium

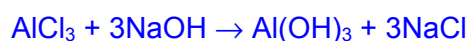
C silicon

B aluminium

D phosphorus

Answer: B

Based on the information provided, the only possible Period 3 element for **J** is Al.



Al is a solid at room temperature.

The oxide of Al is soluble in hydrochloric acid.



30 The use of the Data Booklet is relevant to this question.

The colours of various vanadium ions in aqueous solution are given in the table below.

Oxidation state	V	IV	III	II
Ion	VO_2^+	VO^{2+}	V^{3+}	V^{2+}
Colour	yellow	blue	green	violet

What is the final colour of the solution when excess zinc powder is added to an acidified solution containing VO_2^+ ions?

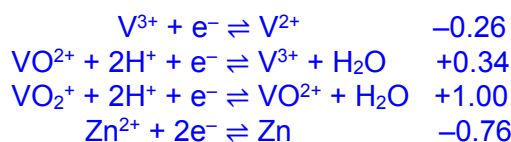
A yellow

C green

B blue

D violet

Answer: D



$$E^\circ_{\text{cell}} = E^\circ_{\text{red}} - E^\circ_{\text{ox}} = E^\circ_{\text{red}} - (-0.76)$$

For reaction to be spontaneous, $E^\circ_{\text{cell}} > 0$ V. In order for E°_{cell} to be > 0 , E°_{red} must be > -0.76 . Thus, VO_2^+ will be eventually reduced to V^{2+} .

End of Paper